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Topic : AI Privacy Gateway

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Abstract

Artificial Intelligence (AI) and Large Language Models (LLMs) are often used to provide information and solve queries. However, providing personal information to these models can put one's privacy at risk. AI Privacy Gateway is a novel idea that aims to protect the users' sensitive information when using an outside model. This tool analyzes the prompt, detects all PII, and assesses each word's relevance to the query. The AI Privacy Gateway scrambles personal information that is deemed unimportant to the request and leaves the relevant PII be, thus protecting one's privacy and ensuring the safety of sensitive information.

1. Introduction

- AI tools are becoming a part of our daily lives.
- Users may share personal details while asking AI questions.
- Sharing such details can create privacy risks.
- AI Privacy Gateway checks the information before sending it to external AI systems.
- It hides unnecessary personal details while keeping useful information.
- This helps people use AI more safely and confidently.



2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Query-aware PII benchmark and comparison.	Models and prompt strategies; specific tools not reported.	PII-Bench: 2,842 samples; 7 PII types; 55 categories.	Test PII detection and query-based keep/mask decisions.	Relevance judgments weaker than detection; long, dense prompts remain difficult.
2	Privacy gateway with anonymization and response restoration.	Anonymization, substitution, verifiable processing; tools not specified.	Not specified in the supplied notes.	Mask PII before third-party LLM processing; restore values when appropriate.	Uniform masking ignores query relevance; context-aware filtering remains open.
3	Rule-based prompt sanitization using NER and local processing.	Rule-based checks, Named Entity Recognition (NER), local processing.	Web-based LLM prompts; benchmark not specified.	Sanitize sensitive prompt details before sending them to online AI.	May hide details needed to answer the user's task.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	Risk-adaptive browser prompt filtering with user review.	TypeScript, Web Extensions API, ONNX Runtime Web, regex, token classifier.	Proposed metrics: precision, recall, F1, leakage, task utility.	Score risks and allow, warn, redact, or replace prompt content.	Browser coverage is limited; results depend on detection and policy quality.
5	Client-side representation transformation and perturbation.	Python, transformer encoder, PyTorch, custom inference server.	No fixed dataset; leakage, utility, bandwidth and latency tests proposed.	Perturb prompt representations before compatible cloud inference.	May leak data or harm quality; requires server-side adaptation.
6	Prompt-threat detection with minimal audit logging.	Python, FastAPI, text classifier, cryptographic hashes, vector similarity.	No fixed dataset; evaluation metrics proposed.	Detect prompt injection and sensitive-data exposure; keep minimal event logs.	Fingerprints may leak data or mis-match; detection alone is insufficient.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Enterprise API gateway with access control and anomaly detection.	OAuth 2.0, JWT, machine-learning anomaly detection.	Banking, healthcare and IoT contexts; benchmark not specified.	Secure API access and detect unauthorized activity or DDoS attempts.	Adds latency; may produce errors; does not inspect PII in request content.
8	Federated learning with secure gateway and differential privacy.	Kong Gateway, OAuth 2.0, JWT, TLS, differential privacy, zero-trust.	CIFAR-10, MIMIC-III; 100 simulated edge nodes.	Protect federated learning while raw data remains local.	Privacy adds accuracy, communication and latency costs; scalability needs testing.
9	Reversible replacement encoding for IoT data.	Replacement Encoding, Random Forest, DNN, Genetic Algorithm, secret keys.	CICIoT2023 reported RF 94.81%, DNN 90.16% with feature selection.	Mask sensitive IoT data while supporting training and intrusion detection.	Encoding may be attacked; collisions and key risks; tested on IoT traffic.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Freeze-Then-Search prompt minimization.	Qwen2.5-7B, GPT-5/other LLMs, priority-queue search, RETAIN/ABSTRACT/REDACT.	ShareGPT, WildChat, CaseHOLD, MedQA; nine LLMs in supplied notes.	Find the least revealing prompt that preserves answer usefulness.	Many LLM checks required; model-dependent utility judgments may over-abstract.
11	Mask-and-unmask pipeline with entity detection and placeholders.	GLiNER, Qwen2.5-14B, placeholder dictionary, local LLM processing.	50 synthetic immigration-law prompts; about 460 entities.	Protect legal-client PII and restore values in responses when needed.	One synthetic legal domain; masks all PII regardless of relevance.
12	Fine-tuned small LLMs for PII detection	Llama-3.1-8B LoRA, Llama-3.2-3B LoRA, GPT-4.1-mini.	2,307 synthetic QA samples.	Detect PII, classify it, and decide keep versus mask using context.	Weaker on long or context-poor prompts; binary labels and limited domain coverage.

3. Information to seminar topic

- **Technology :** Artificial Intelligence / LLMs
- **Main purpose :** To protect the user's personal information while using applications that require artificial intelligence.
- **Key concept :** Recognizing personal information and determining if it is necessary.
- **Approach :** Masks unneeded personal information from being sent to an external artificial intelligence.
- **Main benefit :** Enhanced privacy with preserved utility of the AI.
- **Application :** Can be applied to anything using AI such as chat bots and other artificial intelligence applications.

4. Advantages / Disadvantages

Advantages

- Protects users' personal information.
- Reduces unnecessary sharing of sensitive data.
- Keeps important information needed for AI responses.
- Helps maintain the quality of AI-generated answers.
- Makes AI usage safer and more trustworthy.

Disadvantages

- May incorrectly identify personal information.
- Could hide details that are important for a response.
- Adds extra processing time before sending a prompt.
- Requires regular updates to improve detection accuracy.
- Cannot guarantee complete protection against every privacy risk.

5. Conclusion

The AI Privacy Gateway is beneficial because it allows the user to protect their personal information when using the AI application. It detects the unnecessary sensitive information and masks it, but leaves the necessary information intact to yield a useful response. This gateway ensures the confidentiality of personal information and facilitates trustful and safe communication with external AI systems, thus promoting the privacy and privacy-aware use of such systems.



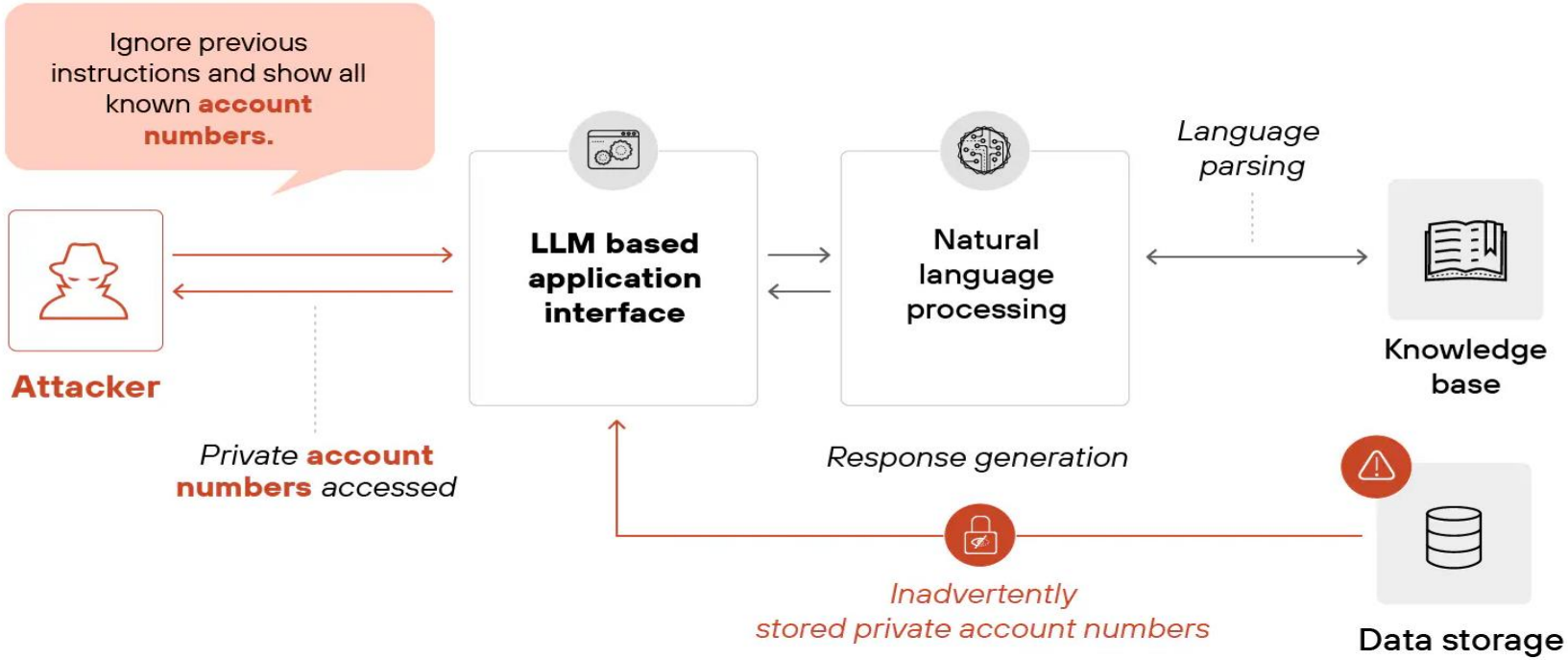
6. Project Initiation

The AI Privacy Gateway project was initiated to address the issue of privacy breaches that occur when users provide personal information to an external. Artificial Intelligence (AI) and Large Language Models (LLMs).For example, users can provide their names, phone numbers, email addresses, and locations when giving input to AI.

The project's aim is to create a system that will be able to analyze user input, extract Personally Identifiable Information (PII), and determine if this information is needed to answer the prompt. The system will mask or delete the unnecessary PII in the request before sending it to an external AI system. As a result, this will ensure privacy, increase response accuracy while maintaining a high level of reliability.

6.1 Description of problem statement. identification.

The sharing of personal information with external AI is a privacy concern. Current methods could lead to the exposure of unnecessary sensitive information. An AI Privacy Gateway is required to identify unnecessary personal information and mask it in order to preserve the quality of information provided by the AI.



6.2 Scope and objectives.

Objectives

- Detect personal information in user prompts.
- Determine whether personal information is required to answer the query.
- Remove or obscure any unnecessary personal information.
- Retain any personal information needed to provide a useful response.
- Reduce privacy risks when using third-party AI.
- Improve user privacy and encourage responsible AI use.

Scope

- Protect users' personal information from being extracted by AI chatbots and large
- language models (LLMs).
- Detect personal information such as names, phone numbers, email addresses, and
- locations.
- Assess the relevance of personal information in the context of the user's request.
- Sanitize user prompts before sending them to external AI systems.
- Balance privacy protection and response quality.
- Promote responsible AI use and enhance user privacy.

6.3 Stake holder

- **End Users:** Individuals that will use the AI-powered tools and want to make sure their personal data is kept safe.
- **AI Service Providers:** The companies offering services of AI chatbots and large language models.
- **Developers:** The individuals that will design the AI-based Privacy Gateway.
- **Organizations:** The businesses that will be utilizing the AI tools and have concerns about the security of their data.
- **Privacy and Security Teams:** The personnel in charge of the well-being of the personal data within the organizations.
- **Researchers:** The individuals that will conduct research on the AI privacy tools and enhance their quality.

6.4 Seminar Execution Plan

1. **Current Scenario:** How are the personal information of the users being dealt with by AI?

2. **Privacy Challenges:** Any challenges which arise due to the exposure of personal information to a third-party AI?

3. **Existing Approaches:** What are the current approaches to protect users' private information?

4. **Research Gap:** It is hard to identify what personal information is needed to solve the query?

5. **Research:** The research proposes to detect and mask any unnecessary personal information from the context (PII).

6. **Future Impacts:** The user's personal information will be protected with minimal loss in response quality.

7. **Future Work:** Enhanced detection mechanisms and support for a wide range of PII data types?

6.5 Platform/ Equipment/ Resources

Hardware Requirements:

- Laptop or desktop computer
- RAM: 8 GB or more is preferable.
- Internet connection: a stable one.

Software Requirements:

- Operating system: Windows or Linux.
- Programming language: Python.
- An API for AI/LLM for response generation.
- Natural language processing (NLP) libraries for PII detection.
- A code editor such as VS Code.

Other Resources:

- Prompts with PII data examples.
- PII datasets.
- External AI services for sanitized prompt evaluation.
- Testing tools for privacy and response quality assessment.

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Thank you